

Carbon Emission Dataset - Exploratory Data Analysis Report

Advanced Python Project - Second Semester 2025/2026

This report summarizes the EDA, cleaning, visualization, and statistical analysis performed on the Carbon emission dataset with Industry sectors.

The project requirement asks students to use Pandas, NumPy, Seaborn, and Matplotlib to clean, explore, visualize, and interpret carbon emission data, and to submit a notebook plus a PDF report.

1. Dataset Overview

The dataset contains **18,250** rows and **22** columns. Each row represents a daily record for a company. The target variable is **Carbon_Emission_tCO2e_TARGET**.

There were **0** missing values and **0** duplicated rows before cleaning. After cleaning, the total missing values are **0**. The Date column was converted to datetime format.

2. Basic Statistics

The table below shows mean, median, maximum, and minimum values for the main numerical variables required in the assignment.

Metric	mean	median	max	min
Total_Energy_Consumption_kWh	124776.987	124728.53	199988.91	50008.73
Renewable_Energy_Consumption_kWh	43759.097	38888.005	119005.26	5041.2
NonRenewable_Energy_Consumption_kWh	81017.89	77007.82	179431.02	20483.26
Production_Output_Units	5538.964	5556.4	9999.78	1000.47
Supply_Chain_Transport_km	2734.814	2735.02	4999.89	500.38

3. Exploratory Visual Analysis

The target distribution is slightly right-skewed. The skewness value is 0.666 and the kurtosis value is -0.004.

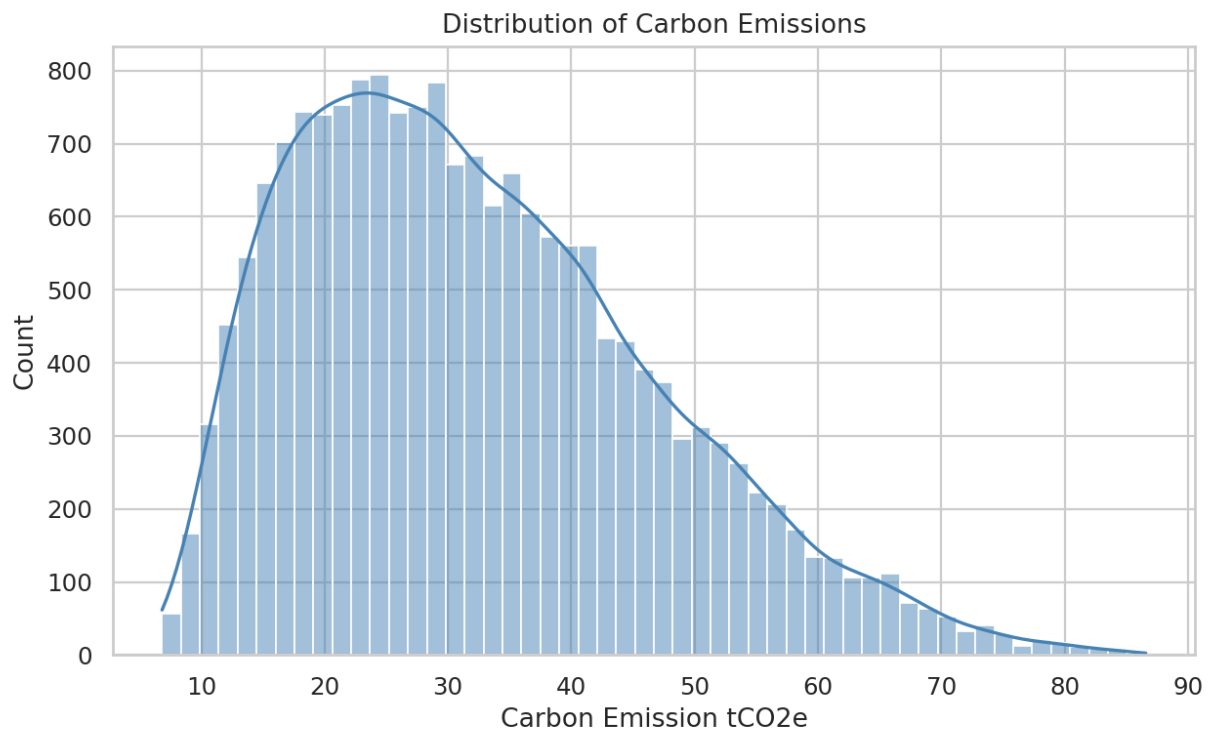


Figure 1: Distribution of Carbon Emissions.

Energy consumption over time is stable at the daily average level. Total energy is consistently higher than non-renewable and renewable components.

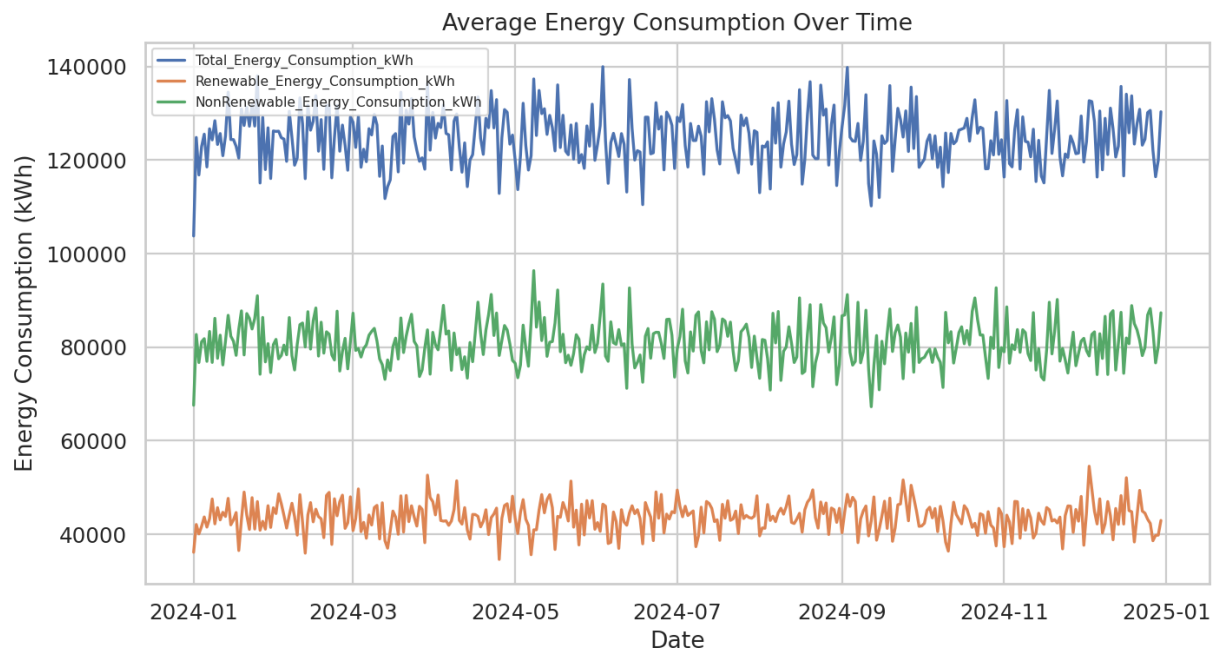


Figure 2: Average energy consumption over time.

4. Bivariate and Multivariate Analysis

The strongest positive relationships with carbon emissions are shown in the correlation analysis. The most important variable is non-renewable energy consumption, followed by carbon tax and total energy consumption.

Metric	Correlation with Target
NonRenewable_Energy_Consumption_kWh	0.937
Carbon_Tax_USD	0.856
Total_Energy_Consumption_kWh	0.778
Energy_Cost_USD	0.626
Expected_Renewable_Share_Percent	-0.477
Renewable_Energy_Consumption_kWh	0.07
Expected_Carbon_Reduction_Percent	0.01
Raw_Material_Usage_kg	0.01

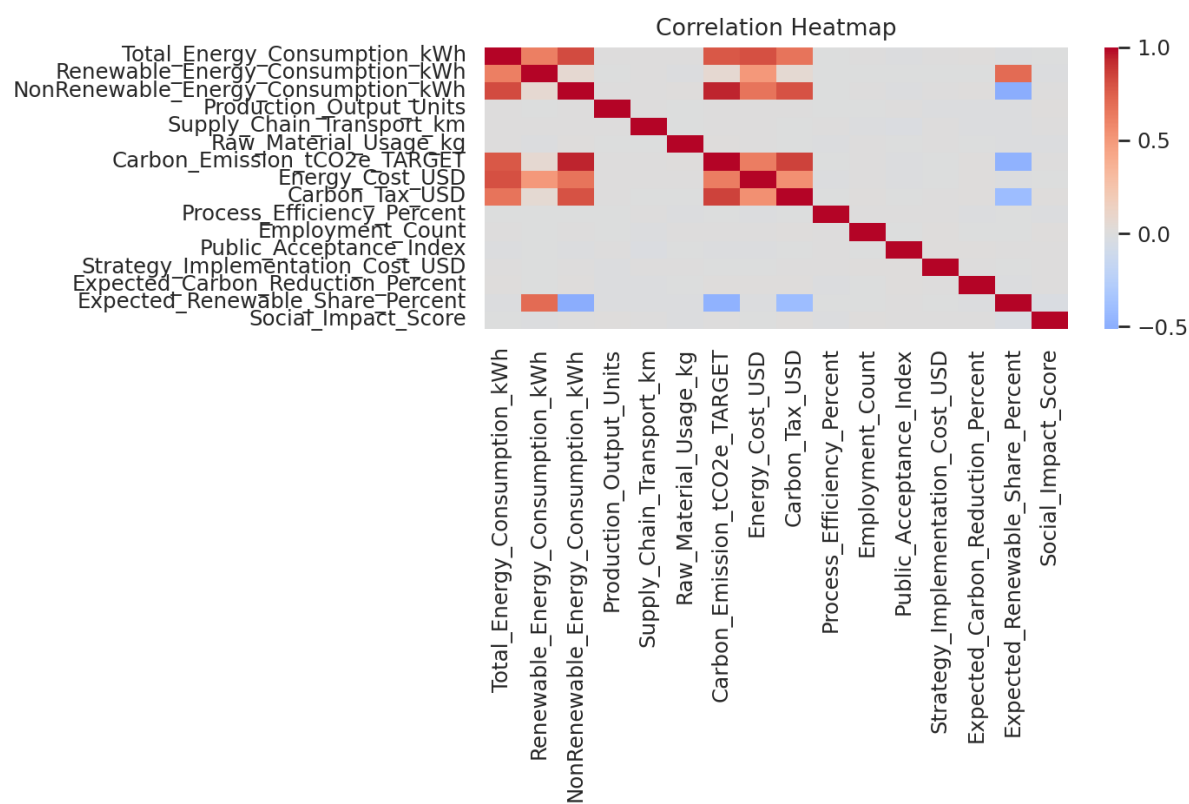


Figure 3: Correlation heatmap of numerical variables.

The box plot by industry sectors shows very similar emission distributions across industry groups, with only small differences in the mean values.

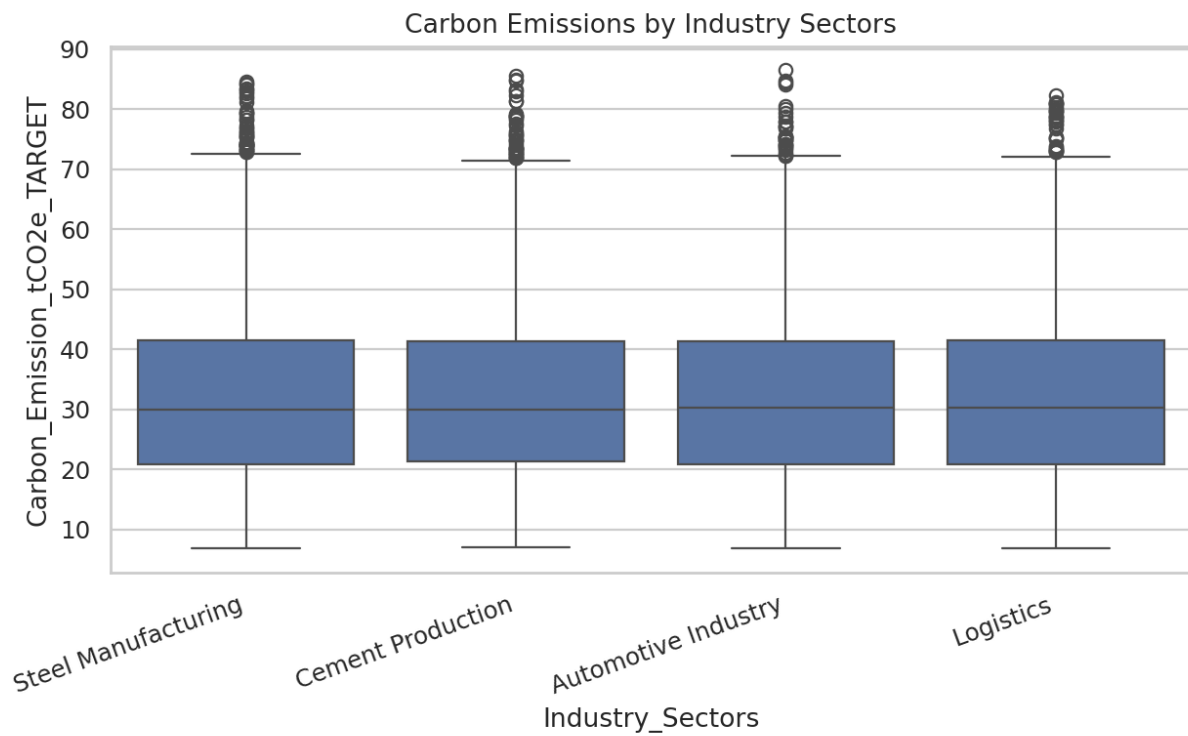


Figure 4: Carbon emissions grouped by industry sectors.

Industry-based mean emissions:

Metric	count	mean	median
Automotive Industry	4562.0	32.404	30.304
Cement Production	4563.0	32.361	30.02
Logistics	4562.0	32.353	30.347
Steel Manufacturing	4563.0	32.342	29.981

Sector-based mean emissions:

Metric	count	mean	median
IT	1460.0	32.496	30.508
Manufacturing	5110.0	32.45	30.291
Agriculture	4380.0	32.41	30.298
Transport	3650.0	32.338	29.887
Energy	3650.0	32.167	29.905

5. Time Series Exploration

Daily average emissions were plotted over time and compared using a 30-day moving average. The last 30-day average changed by 1.51% compared with the first 30-day average.

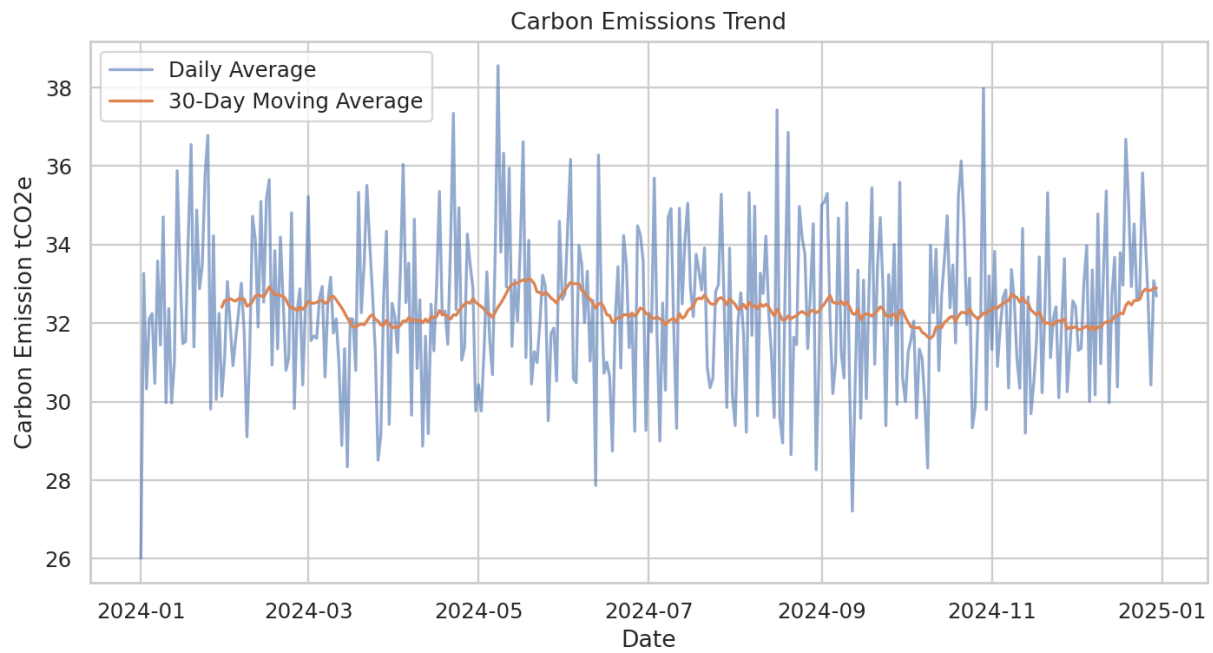


Figure 5: Carbon emissions trend with 30-day moving average.

The time series decomposition separates the data into observed values, trend, seasonal component, and residuals. The seasonal signal is weak because the monthly average range is only 1.016 tCO₂e.

Time Series Decomposition

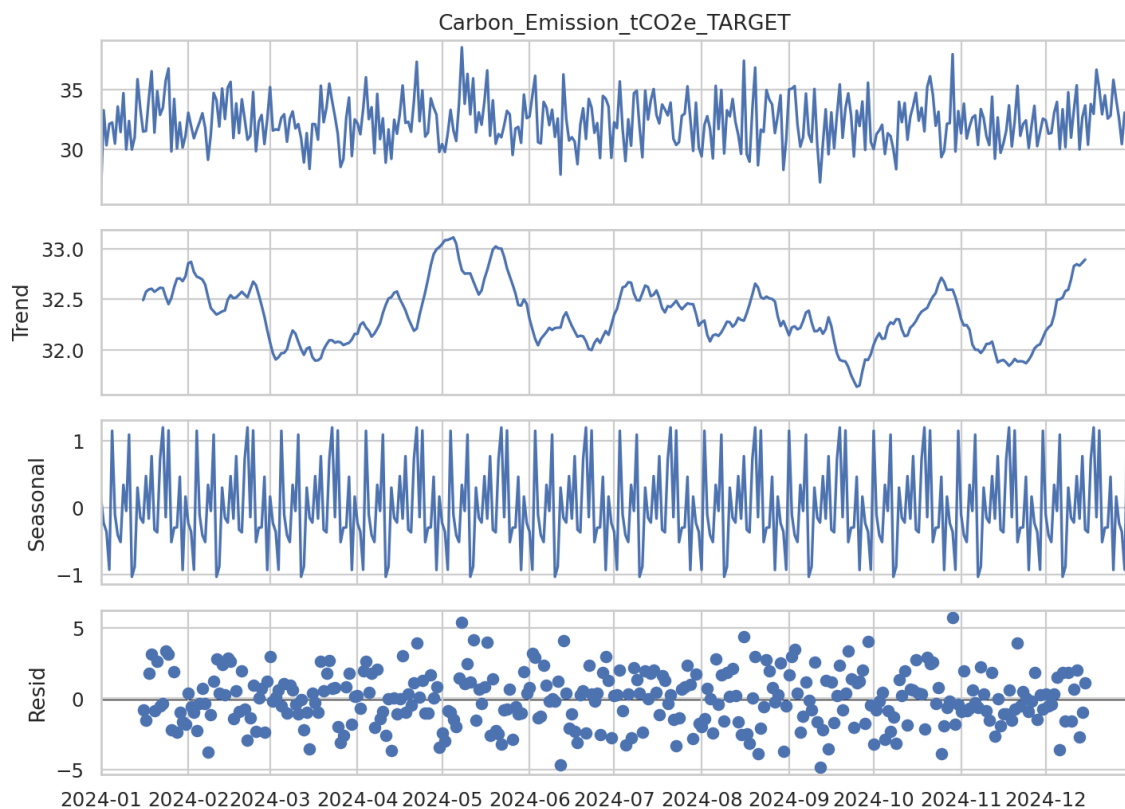


Figure 6: Time series decomposition of carbon emissions.

6. Answers to Advanced Questions

Q1: Which emission sources have highest variability?

The highest variability by coefficient of variation is **Renewable_Energy_Consumption_kWh** with CV = 0.555.

Q2: Are emissions normally distributed?

The emission distribution is not perfectly normal. It is right-skewed with skewness = 0.666.

Q3: Which variables are strongly correlated?

The strongest correlations with emissions are non-renewable energy consumption (0.937), carbon tax (0.856), total energy consumption (0.778), and energy cost (0.626).

Q4: Is there a trend in emissions?

There is no strong long-term trend. The first 30 days average and last 30 days average differ by only 1.51%.

Q5: Do seasonal patterns exist?

Seasonal patterns are weak. The monthly average range is 1.016 tCO₂e, so the seasonal effect is small in this dataset.

7. Important Note

The project description asks for emissions vs temperature and temperature trend over time. The uploaded dataset does not contain a temperature column. Therefore, the notebook includes conditional code that searches for a temperature column and prints a note if it does not exist.

8. Conclusion

The analysis indicates that carbon emissions are mainly driven by non-renewable energy consumption, total energy use, and related carbon tax values. Renewable energy share is negatively associated with emissions, which supports the idea that increasing renewable energy can help reduce carbon footprint.